

The "Why"

- It works on my machine!



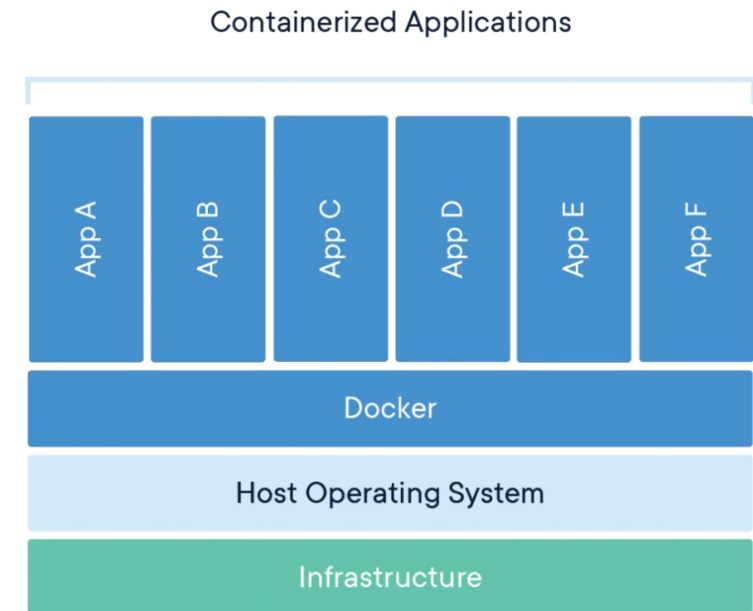
The Problem: You finish your project. It works perfectly on your Windows laptop with Python 3.14. You send it to a colleague who has a Mac and Python 3.10. **It crashes.**

The AI Specific Problem: AI libraries (PyTorch, TensorFlow, OpenCV) are very sensitive to versions. If one library is off by a tiny version number, the whole model might fail.

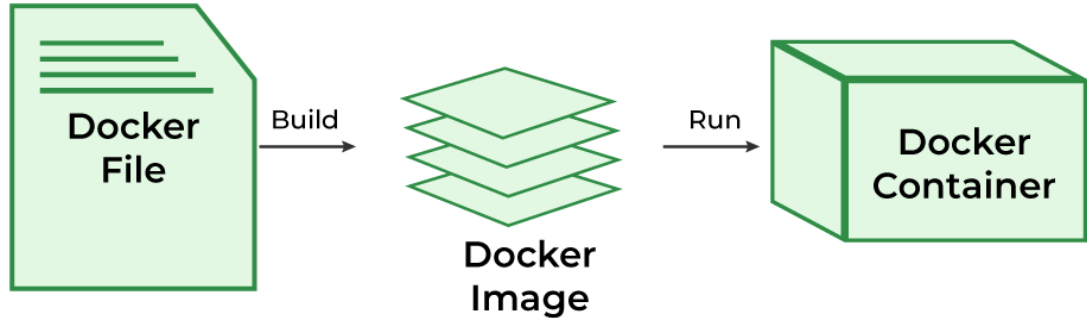
The Solution: Docker. Instead of just sending the code, you send a "Container" that includes the code, the Python version, and every single library installed exactly as you like it.

The "What" - The Shipping Container

- **Analogy:** Before shipping containers, loading a boat was a mess. Now, every container is a standard size. You don't care if there are cars or coffee inside; the crane just picks it up.
- **Docker Image:** This is your "Recipe" (The Blueprint).
- **Docker Container:** This is the "Actual Cake" (The running instance of your code).
- **Docker Hub:** The "Global Warehouse" where you store your images.



The "How" - The Dockerfile



- To create a container, you write a **Dockerfile**. Think of this as a script that:
- Starts a fresh, clean computer (the Base Image).
- Installs Python.
- Copies your code into it.
- Runs your script.